

SLO-05: SLOs as Code

Series: SLO — Service Level Objectives | **Notebook:** 5 of 5 | **Created:** June 2026
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Overview

A validated SLO that lives only in the UI is one accidental click from gone, and impossible to reproduce across environments. This notebook covers promoting SLOs into version control: the `builtin:monitoring.slo` schema, the `dynatrace_slo_v2` Terraform resource, the Monaco alternative, and the API path — with a deliberate emphasis on verifying field names at the source rather than copying a payload that may be stale.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS Gen3 with the SLO app

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Tooling	Terraform with the <code>dynatrace-oss/dynatrace</code> provider, or Monaco
Auth	Platform Token or OAuth client per the AUTOM-04 auth-scheme guidance
Prior reading	SLO-02 (the SLI query you will codify), AUTOM-04 / AUTOM-07 (provider auth, CI/CD)

1. Why Version SLOs

SLOs are configuration, and configuration that matters belongs in source control:

- **Reproducible across environments** — the same SLO in dev, staging, and prod from one definition.
- **Reviewable** — a target change goes through a PR, not a quiet UI edit.
- **Recoverable** — an accidental deletion is a `terraform apply` away from restored.

The workflow mirrors AIOPS-02's stance on detectors: **prototype in the app, promote to code once it matters.**



2. The Schema and Terraform Resource

Surface	Identifier
Settings 2.0 schema	<code>builtin:monitoring.slo</code>
Terraform resource	<code>dynatrace_slo_v2</code>
Monaco	<code>builtin:monitoring.slo</code> schema via <code>--settings-schema</code>

A gap to plan around: the `dynatrace_slo_v2` resource's SLI field (`metric_expression`) takes a **classic metric-selector expression** — e.g. `100*(builtin:service.requestCount.server:splitBy())/(...)` — not the DQL query with an `qli` field that a Custom SLO in the app uses (SLO-01 §3, SLO-02). If your SLI is built from a pre-aggregated metric, this maps over directly, and Dynatrace's own docs recommend metric-based SLOs for performance and cost where a suitable metric exists. If your SLI is genuinely DQL-only — a span-ratio or log/bizevent-derived `qli`, as in SLO-02's latency and custom examples — there is no documented Terraform field for it today; codify those via the export-a-known-good-object path in Section 5 instead of hand-writing HCL.

Verify the exact argument names against the provider docs before you write HCL. The provider schema evolves between releases, and copying a stale payload is precisely how the field-authored alerting document ended up with an SLO API body that would not apply. The example below reflects the resource's documented schema as fetched at source on 07/01/2026 — treat the [resource docs](#) as authoritative going forward, since the schema can still change between provider releases.

```
# dynatrace_slo_v2 – schema verified at the provider registry 07/01/2026
# Metric-based SLI (classic metric-selector syntax) – see the gap callout in
Section 2
# regarding DQL-only SLIs, which this resource does not accept directly.
resource "dynatrace_slo_v2" "web_availability_30d" {
  name          = "Web Service Availability – 30d"
  custom_description = "Request success ratio for the critical web service,
rolling 30 days"
  enabled       = true
  evaluation_type = "AGGREGATE"

  # rolling 30-day window
  evaluation_window = "-30d"

  # entitySelector syntax (not a DQL filter) – scopes the SLO to web services
  filter = "type(SERVICE),serviceType(WEB_SERVICE,WEB_REQUEST_SERVICE)"

  # classic metric-selector expression – good / total as a percentage
  metric_expression = "100*
(builtin:service.successCount:splitBy())/(builtin:service.requestCount:splitB
y())"
  metric_name      = "web_availability_30d"

  target_success = 99.5
  target_warning = 99.9

  # required block – burn-rate visualization + optional fast-burn threshold
  (SLO-04)
  error_budget_burn_rate {
    burn_rate_visualization_enabled = true
    fast_burn_threshold             = 14
  }
}
```

3. Terraform Worked Example — Notes

A few things the example encodes:

- **target_success** and **target_warning** are the goal and the early-warning line from SLO-01 §4 (note the argument names — not the shorter **target** / **warning** an earlier draft of this notebook assumed). Warning above target surfaces "getting close" before an actual breach.

- `metric_expression` is a classic metric-selector string, not a DQL query — see the Section 2 gap callout. It is the codified form of a good/total ratio built from a pre-aggregated metric, analogous in intent to the SLO-02 DQL SLI but not the same syntax.
- `filter` uses entitySelector syntax (`type(SERVICE),serviceType(...)`), not a DQL filter clause and not quoted the way a DQL string literal would be. An unscoped SLO measures the whole environment, which is rarely what you want.
- `error_budget_burn_rate` **is a required block**, not optional — even if you don't need the visualization, the block itself must be present. `fast_burn_threshold` is optional within it and pairs with the SLO-04 burn-rate alerting recipe.
- **State and auth** follow the same rules as every other Dynatrace Terraform resource — see AUTOM-04 for Platform-Token vs classic-token routing and AUTOM-09 for state-backend setup.

4. Monaco Alternative

If your shop standardises on Monaco rather than Terraform, the same SLO is a `builtin:monitoring.slo` settings object:

```
configs:
  - id: web-availability-30d
    type:
      settings:
        schema: builtin:monitoring.slo
        scope: environment
      config:
        template: web-availability-30d.json
        skip: false
```

The JSON template carries the SLI, target, warning, and window — and, unlike the Terraform resource, the Settings API payload behind a DQL-based Custom SLO (SLO-01 §3) is a first-class fit here, since Monaco pushes whatever JSON shape the schema accepts rather than mapping through the narrower Terraform resource attributes. As with Terraform, validate the SLI query in a notebook first, then export the working SLO from the app with `monaco download` to capture the exact current JSON shape rather than hand-writing it.

5. API and CI/CD

For direct API automation, use the **current SLO endpoint**, not the deprecated v1 SLO API. Rather than reproduce a request body that may drift, generate it from a working SLO:

1. Create and validate one SLO in the app.
2. Read it back via the API (or `monaco download` / `terraform-provider-dynatrace - export`) to capture the exact current schema.
3. Template that shape for the rest.

This "export a known-good object" approach is the antidote to stale-payload errors, and it is also the path of least resistance for a DQL-only SLI, since the Terraform resource's `metric_expression` field cannot carry one (Section 2). For the full CI/CD pattern — plan on PR, gated apply, three-way validation — see AUTOM-07 (pipelines) and AUTOM-96 (GitHub Actions LAB); the SLO resource slots into exactly the same pipeline as any other Dynatrace config.

Sources: [Service-Level Objectives \(DT docs\)](#), [dynatrace_slo_v2 resource \(Dynatrace provider docs\)](#), [Create service-level objectives \(DT docs\)](#), [Monaco configuration \(DT docs\)](#). Terraform schema (`builtin:monitoring.slo` , `evaluation_window` , `target_success` / `target_warning` , required `error_budget_burn_rate` block) verified at the provider registry 07/01/2026. **Derived:** the metric-expression-vs-DQL-sli gap in Section 2 combines the provider's resource schema with the Create-SLO docs' DQL `sli` -field description — neither source states the gap explicitly.

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